

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
OPEN TEST – II
PAPER –1
TEST DATE: 20-03-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. D

Sol. $F_0 \cos \theta = F$... (i)

$$F_0 \sin \theta = \frac{\lambda \ell g}{2} \quad \dots (ii)$$

Solving (i) and (ii), we get

$$F = \frac{\lambda \ell g}{2} \cot \theta \quad \dots (iii)$$

Let the radius of curvature of the rope at the lowest point be r

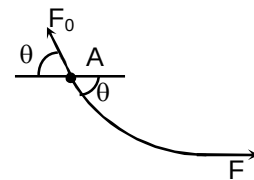
$$F d\theta = \lambda r d\theta g$$

$$F = \lambda g r$$

$$r = \frac{F}{\lambda g} = \frac{\ell}{2} \cot \theta \quad \text{(from iii)}$$

$$r = \frac{12}{2} \times \frac{4}{3} = 8 \text{ m}$$

$$r = 8 \text{ m}$$



2. B

Sol. $\Delta Q_{AB} = n C_V \Delta T = 1 \times \frac{3R}{2} \times 900 = 1350R$

In the process BC,

$$TV = \text{constant}$$

$$PV^2 = \text{constant}$$

Molar heat capacity of the gas in the process BC,

$$C = C_V + \frac{R}{(1-x)}$$

$$C = \frac{3R}{2} + \frac{R}{(1-2)} = \frac{3R}{2} - R = \frac{R}{2}$$

$$C = \frac{R}{2}$$

$$\Delta Q_{BC} = nC\Delta T = 1 \times \frac{R}{2} \times (600 - 1200) = -300R$$

$$\Delta Q_{CA} = nC_p\Delta T = 1 \times \frac{5R}{2} \times (300 - 600) = -750R$$

$$\Delta W_{\text{cycle}} = \Delta Q_{\text{cycle}} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA}$$

$$\Delta W_{\text{cycle}} = 1350R - 300R - 750R = 300R$$

$$\eta = \frac{\Delta W_{\text{cycle}}}{\Delta Q_{\text{supplied}}} \times 100 = \frac{300R}{1350R} \times 100$$

$$\eta = \frac{200}{9} = 22.2\%$$

3. B

Sol.

$$Z_1 = R_1 + X_L i = 8 + 6i$$

$$Z_2 = R_2 - X_C i = 3 - 4i$$

$$\text{Now, } \frac{1}{Z'} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\frac{1}{Z'} = \frac{1}{(8 + 6i)} + \frac{1}{(3 - 4i)}$$

$$= \frac{(8 - 6i)}{100} + \frac{(3 + 4i)}{25} = \frac{20 + 10i}{100}$$

$$\frac{1}{Z'} = \frac{(2 + i)}{10} \Rightarrow Z' = \frac{10}{(2 + i)}$$

$$Z' = 2(2 - i)$$

$$Z' = (4 - 2i) \quad \dots(i)$$

$$Z_3 = R_3 + X_L' i = (2 + 10i) \quad \dots(ii)$$

Hence, net impedance of the circuit is

$$Z = Z' + Z_3 = (4 - 2i) + (2 + 10i)$$

$$Z = 6 + 8i$$

$$|Z| = \sqrt{(6)^2 + (8)^2} = 10 \Omega$$

$$\text{Hence, } I = \frac{80}{|Z|} = \frac{80}{10} = 8 \text{ amp}$$

4. C

Sol.

$$E_1 = \frac{hc}{\lambda_1} = \frac{1240}{310} = 4 \text{ eV}$$

$$E_2 = \frac{hc}{\lambda_2} = \frac{1240}{400} = 3.1 \text{ eV}$$

$$\text{Also, } \frac{k_1}{k_2} = 4 \Rightarrow k_1 = 4k_2$$

$$\text{Now, } E_1 = k_1 + \phi$$

$$k_1 + \phi = 4 \quad \dots(i)$$

$$E_2 = k_2 + \phi$$

$$k_2 + \phi = 3.1 \quad \dots(ii)$$

Solving (i) and (ii), we get

$$k_1 - k_2 = 0.9$$

$$4k_2 - k_2 = 0.9$$

$$3k_2 = 0.9$$

$$\Rightarrow k_2 = 0.3 \text{ eV}$$

$$k_1 = 1.2 \text{ eV}$$

from equation (i),

$$k_1 + \phi = 4$$

$$\phi = 4 - k_1 = 4 - 1.2 = 2.8 \text{ eV}$$

$$\phi = 2.8 \text{ eV}$$

5. A, B, D

Sol. $I_P = 2mR^2 + mR^2 = 3mR^2$

Using conservation of energy,

$$mgR \sin 30^\circ = \frac{1}{2} \times 3mR^2 \omega^2$$

$$\frac{mgR}{2} = \frac{3}{2} mR^2 \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{g}{3R}}$$

...(i)

Now, $\tau_P = I_P \alpha$

$$2m(g + \omega^2 R) \frac{R}{2} \cos 30^\circ = 3mR^2 \alpha$$

$$\left(g + \frac{g}{3}\right) \frac{\sqrt{3}}{2} = 3R \alpha$$

$$\Rightarrow \alpha = \frac{2g}{3\sqrt{3}R}$$

...(ii)

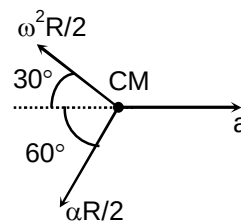
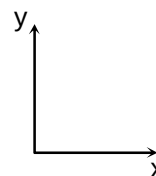
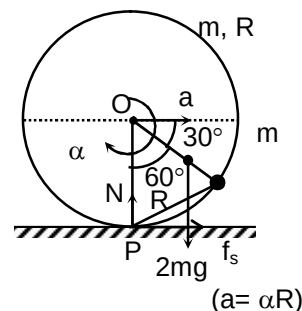
$$a_{CM(x)} = a - \frac{\omega^2 R}{2} \cos 30^\circ - \frac{\alpha R}{2} \cos 60^\circ$$

$$= \alpha R - \omega^2 R \frac{\sqrt{3}}{4} - \frac{\alpha R}{4}$$

$$a_{cm(x)} = \frac{3\alpha R}{4} - \omega^2 R \frac{\sqrt{3}}{4} = \frac{g}{2\sqrt{3}} - \frac{g}{4\sqrt{3}}$$

$$a_{cm(x)} = \frac{g}{4\sqrt{3}}$$

$$f_s = 2ma_{cm(x)} = 2m \times \frac{g}{4\sqrt{3}} = \frac{mg}{2\sqrt{3}}$$



6. A, B, C

Sol. $A_r = \left(\frac{v_2 - v_1}{v_2 + v_1} \right) A_i = \left(\frac{\sqrt{\frac{F}{\mu_2}} - \sqrt{\frac{F}{\mu_1}}}{\sqrt{\frac{F}{\mu_2}} + \sqrt{\frac{F}{\mu_1}}} \right) A_i$

$$A_r = \left(\frac{1 - \sqrt{\frac{\mu_2}{\mu_1}}}{1 + \sqrt{\frac{\mu_2}{\mu_1}}} \right) = \left(\frac{1 - \frac{2}{3}}{1 + \frac{2}{3}} \right) 2.5 = \frac{1}{5} \times 2.5$$

$$A_r = 0.5 \text{ cm}$$

$$A_t = \left(\frac{2v_2}{v_2 + v_1} \right) A_i = \left(\frac{2\sqrt{\frac{F}{\mu_2}}}{\sqrt{\frac{F}{\mu_2}} + \sqrt{\frac{F}{\mu_1}}} \right) A_i = \frac{2A_i}{\left(1 + \sqrt{\frac{\mu_2}{\mu_1}} \right)} = \frac{2 \times 2.5}{\left(1 + \frac{2}{3} \right)}$$

$$A_t = 3 \text{ cm}$$

The percentage of incident power transmitted from the joint

$$\frac{P_t}{P_i} \times 100 = \left(\frac{A_t}{A_i} \right)^2 \frac{v_1}{v_2} \times 100 = \left(\frac{3}{2.5} \right)^2 \times \sqrt{\frac{\mu_2}{\mu_1}} \times 100$$

$$= \frac{36}{25} \times \frac{2}{3} \times 100 = 96\%$$

The percentage of incident power reflected from the joint O.

$$\frac{P_r}{P_i} \times 100 = \left(\frac{A_r}{A_i} \right)^2 \times 100 = \left(\frac{0.5}{2.5} \right)^2 \times 100 = \frac{1}{25} \times 100 = 4\%$$

7. A, C

Sol. $\frac{1}{v} - \frac{1}{-5} = \frac{1}{10}$

$$\Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{1}{5} = \frac{-1}{10}$$

$$v = -10 \text{ cm}, m = \frac{v}{u} = \frac{-10}{-5} = 2$$

$$d = S_1 S_2 = 1 \text{ mm}$$

$$D = 10 + 30 = 40 \text{ cm} = 0.4 \text{ m}$$

$$\text{Fringe width, } \omega = \frac{\lambda D}{d}$$

$$\omega = \frac{5 \times 10^{-7} \times 0.4}{1 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$$

$$\omega = 0.2 \text{ mm}$$

$$\text{Now, } \frac{L}{d} = \frac{30}{10} \Rightarrow L = 3d = 3 \text{ mm}$$

The number of interference bands observed on the screen,

$$n = \frac{L}{\omega} = \frac{3}{0.2} = 15$$

$$n = 15$$

8. A, D

Sol. $I = \frac{Q\omega}{2\pi}, B = \frac{\mu_0 I}{\ell} = \frac{\mu_0 Q\omega}{2\pi\ell}$

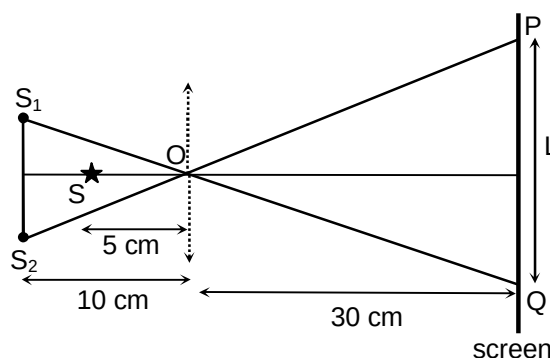
$$E = \frac{R}{2} \frac{dB}{dt} = \frac{R}{2} \frac{\mu_0 Q\alpha}{2\pi\ell} = \frac{\mu_0 Q\alpha}{4\pi\ell} \quad \dots(i)$$

$$mg \sin \theta - f_s = ma \quad \dots(ii)$$

$$f_s R - QE = mR^2 \alpha$$

$$f_s - QE = ma \quad (\because a = \alpha R)$$

$$f_s - \frac{\mu_0 Q^2 a}{4\pi\ell} = ma$$



$$\Rightarrow f_s - \frac{ma}{4} = ma$$

$$f_s = \frac{5ma}{4} \quad \dots(iii)$$

Solving (ii) and (iii), we get

$$mg \sin \theta = \frac{9ma}{4} \Rightarrow a = \frac{4g \sin \theta}{9}$$

$$\text{Hence, } f_s = \frac{5mg \sin \theta}{9}$$

9. B, C

Sol. Using conservation of energy

$$\frac{1}{2} \times \frac{5m\ell^2 \omega^2}{4} = \frac{3mg\ell}{\sqrt{3}}$$

$$\omega = \sqrt{\frac{8\sqrt{3}g}{5\ell}} \quad \dots(i)$$

$$\text{Magnetic moment of the loop, } M = \frac{\sqrt{3}}{4} i \ell^2$$

$$\text{Magnetic torque experienced by the loop, } \tau = MB = \frac{\sqrt{3}}{4} Bi \ell^2$$

$$\text{Now, } \int \tau dt = I \omega$$

$$\frac{\sqrt{3}}{4} BL^2 \int i dt = \frac{5m\ell^2 \omega}{4}$$

$$\frac{\sqrt{3}}{4} B \ell^2 Q = \frac{5m\ell^2 \omega}{4}$$

$$\Rightarrow B = \frac{5m\omega}{\sqrt{3}Q} \quad \dots(ii)$$

From (i) and (ii), we get

$$B = \frac{5m}{\sqrt{3}Q} \sqrt{\frac{8\sqrt{3}g}{5\ell}}$$

$$B = \frac{m}{Q} \sqrt{\frac{40g}{\sqrt{3}\ell}}$$

10. A, B, D

Sol. Pitch of the screw gauge, $P = 1 \text{ mm}$

$$\text{Least count} = \frac{P}{N} = \frac{1}{100} = 0.01 \text{ mm}$$

$$\text{Zero error} = -6 \times \text{L.C.} = -6 \times 0.01 \text{ mm} = -0.06 \text{ mm}$$

The diameter of the wire,

$$d = 3 \text{ mm} + 58 \times \text{L.C.} + 6 \times \text{L.C.}$$

$$d = 3 \text{ mm} + 58 \times 0.01 \text{ mm} + 6 \times 0.01 \text{ mm}$$

$$d = 3 \text{ mm} + 0.58 \text{ mm} + 0.06 \text{ mm}$$

$$d = 3.64 \text{ mm}$$

Section – B

11. 16

Sol. Initially the electric potential at any point of the circular plate due to the uniformly charged hemispherical shell is

$$V = \frac{Q}{8\pi\epsilon_0 R} = \frac{\sigma R}{2\epsilon_0}$$

Now, using conservation of energy

$$K_{\max} = U_i - U_f$$

$$K_{\max} = QV - 0$$

$$K_{\max} = \sigma\pi R^2 \left(\frac{\sigma R}{2\epsilon_0} \right)$$

$$K_{\max} = \left(\frac{\sigma^2 \pi R^3}{2\epsilon_0} \right)$$

 Hence, $n = 16$

12. 9

 Sol. If the pressure of the gas is P when the marble is in equilibrium

$$PS = P_0 S + mg \quad \dots(i)$$

 When the marble is getting at displacement y below the mean position in the tube,

$$dP = -\left(\frac{\gamma P}{V} \right) (-Sy)$$

$$dP = \left(\frac{5PS}{3V} \right) y$$

$$dP = \frac{5}{3V} (P_0 S + mg)y \quad \dots(ii)$$

$$\text{Now, } m \frac{d^2 y}{dt^2} = -dPS$$

$$\frac{d^2 y}{dt^2} = \frac{-5S(P_0 S + mg)y}{3mV}$$

Time period of oscillations of the marble,

$$T = 2\pi \sqrt{\frac{3mV}{5S(P_0 S + mg)}}$$

 Hence, $k = 9$

13. 6

Sol. The equation of longitudinal standing wave is

$$y = A \sin kx \cos \omega t$$

$$\text{Now, } dU = \frac{1}{2} \rho v^2 \left(\frac{\partial y}{\partial x} \right)^2 dx$$

$$dU = \frac{1}{2} \rho v^2 (kA \cos kx \cos \omega t)^2 dx$$

$$dU = \frac{1}{2} \rho v^2 k^2 A^2 \cos^2 \omega t \cos^2 kx dx$$

$$\int dU = \frac{1}{2} \rho \omega^2 A^2 \cos^2 \omega t \int_0^{\ell} \cos^2 kx dx$$

$$U = \frac{1}{2} \rho \omega^2 A^2 S \cos^2 \omega t \int_0^\ell \left(\frac{1 + \cos 2kx}{2} \right) dx$$

$$U = \frac{1}{4} \rho \omega^2 A^2 s \ell \cos^2 \omega t$$

$$\text{At } t = \frac{\pi}{6\omega}$$

$$U = \frac{1}{4} \rho \omega^2 A^2 s \ell \cos^2 \left(\frac{\pi}{6} \right)$$

$$U = \frac{3}{16} \rho \omega^2 A^2 s \ell$$

Hence, $\alpha = 6$

Section – C

14. 1.20

15. 1.00

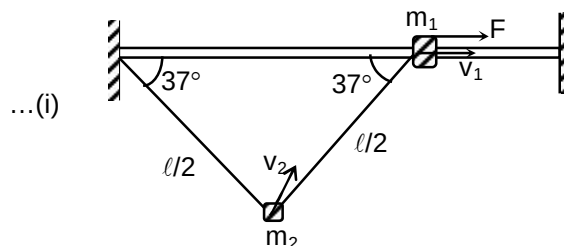
Sol. (for Q.14-15)

$$v_1 \cos \theta = v_2 \sin 2\theta$$

$$v_1 = 2v_2 \sin \theta$$

$$v_1 = 2v_2 \left(\frac{3}{5} \right) \Rightarrow v_2 = \frac{5}{6} v_1$$

Using work energy theorem



$$F(\cos \theta - \cos \theta_0) - m_2 g \frac{\ell}{2} (\sin \theta_0 - \sin \theta) = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$20\ell \left(\frac{4}{5} - \frac{3}{5} \right) - 20 \times \frac{\ell}{2} \left(\frac{4}{5} - \frac{3}{5} \right) = \frac{1}{2} \times 5v_1^2 + \frac{1}{2} \times 2 \times \frac{25}{36} v_1^2$$

$$2(4\ell - 2\ell) = \left(5 + \frac{50}{36} \right) v_1^2 \Rightarrow v_1 = \sqrt{\frac{4 \times 2.3 \times 36}{230}} = 1.2 \text{ m/s}$$

$$v_1 = 1.2 \text{ m/s}$$

$$v_2 = \frac{5}{6} v_1 = 1 \text{ m/s}$$

16. 50.00

17. 18.75

Sol. (for Q. 16-17)

$$\left(\frac{D+x}{D-x} \right)^2 = 9$$

$$\Rightarrow \left(\frac{D+x}{D-x} \right) = 3$$

$$\frac{x}{D} = \frac{2}{4} \Rightarrow x = \frac{D}{2} = \frac{100}{2} = 50 \text{ cm}$$

$x = 50 \text{ cm}$

Focal length of the lens,

$$f = \frac{D^2 - x^2}{4D}$$

$$f = \frac{(100)^2 - (50)^2}{4 \times 100} = \frac{7500}{4 \times 100} = 18.75 \text{ cm}$$

$$f = 18.75 \text{ cm}$$

18. 5.00

19. 6.10

Sol. **(for Q. 18-19)**

$$P_2 \cos \theta = P \quad \dots(i)$$

$$P_2 \sin \theta = P_1 \quad \dots(ii)$$

Solving (i) and (ii), we get

$$P_2^2 = P^2 + P_1^2$$

$$2m_2 k_2 = 2mk + 2m_1 k_1$$

$$10k_2 = 4k + k_1$$

$$10k_2 - k_1 = 4k$$

$$10k_2 - k_1 = 56 \quad \dots(iii)$$

$$\text{Also, } k_1 + k_2 = k - |Q|$$

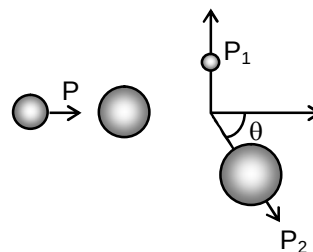
$$k_1 + k_2 = 14 - 2.9$$

$$k_1 + k_2 = 11.1 \quad \dots(iv)$$

Solving (iii) and (iv), we get

$$11k_2 = 67.1 \Rightarrow k_2 = 6.1 \text{ MeV}$$

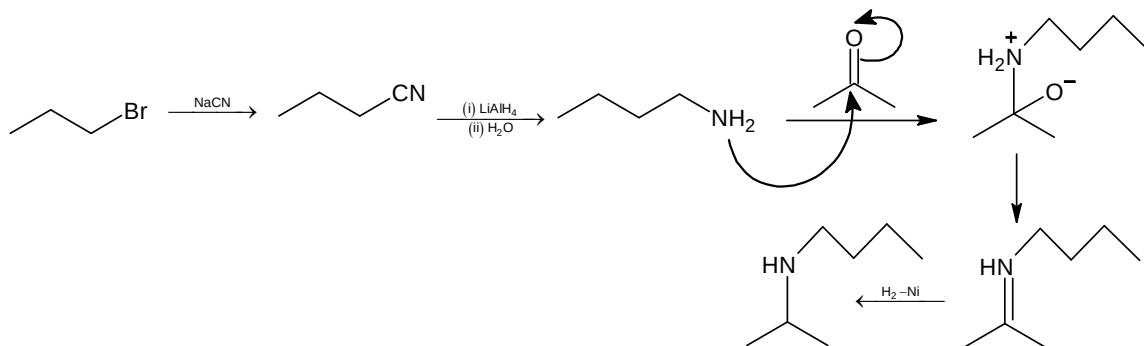
$$k_1 = 5.0 \text{ MeV}$$



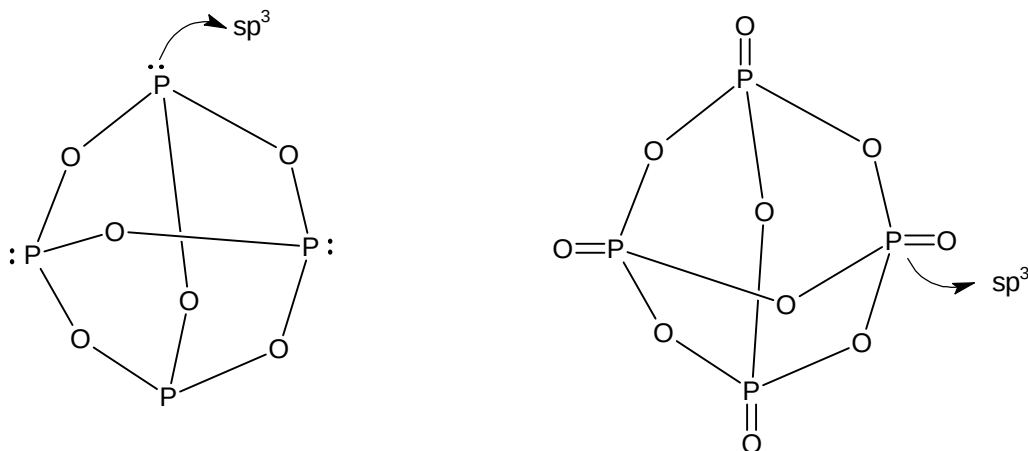
Chemistry**PART – II****Section – A**

20. D
Sol. Compound 3 is trans 1,3-dimethyl cyclohexane.

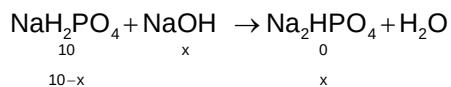
21. A
Sol.



22. A
Sol.



23. B
Sol. $\text{H}_3\text{PO}_4 + \text{NaOH} \rightarrow \text{NaH}_2\text{PO}_4 + \text{H}_2\text{O}$
- Let millimole of NaOH further added are 'x'



$$\text{pH} = 8.2 + \log\left(\frac{x}{10-x}\right)$$

$$\Rightarrow x = 5$$

Therefore total millimoles of NaOH = 10 + 5 = 15

Let volume of NaOH is V_{ml}

$$1 \times V_{\text{ml}} = 15$$

$$V_{\text{ml}} = 15$$

24. A, B, C, D

 Sol. Compound A is Pb_3O_4 , Compound D is PbSO_4 , Compound B is PbO_2 and Compound E is MnO_4^-

$$\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \longrightarrow 2\text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + 2\text{H}_2\text{O}$$

25. B

 Sol.
$$\text{NCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{NH}_3 + 3\text{HOCl}$$

$$\text{PCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{P}(\text{OH})_3 + 3\text{HCl}$$

$$\text{BCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{B}(\text{OH})_3 + 3\text{HCl}$$
 } same pattern of hydrolysis

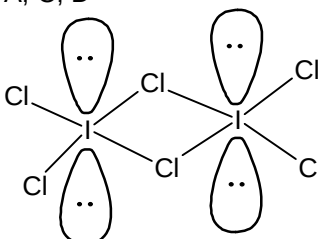
$$\text{NF}_3 + \text{H}_2\text{O} \xrightarrow{\Delta} \text{N}_2\text{O}_3 + \text{HF}$$

$$\quad \quad \quad \downarrow \text{RT}$$

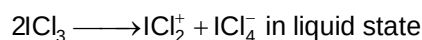
$$\quad \quad \quad \text{No reaction}$$

$$\text{BiCl}_3 + \text{H}_2\text{O} \longrightarrow \text{BiOCl} \downarrow + 2\text{HCl}$$

26. A, C, D

 Sol. 

dimer form of ICl_3 in solid state which is sp^3d^2 hybridized.



27. B, C, D

 Sol. In the version of 1 atm to 50 atm at 200 K.

$$\text{CO}_2(\text{s}) \rightleftharpoons \text{CO}_2(\text{g}) \text{ equilibrium exist.}$$

28. B

 Sol. NH_3 is relatively SFL for $\text{Co}(\text{III})$ so it can't be replaced easily. So, backward reaction is more favoured for NH_3 .

29. B

Sol. Micelle formation is accompanied by the mostly due to increase in solvent entropy increase in overall entropy.

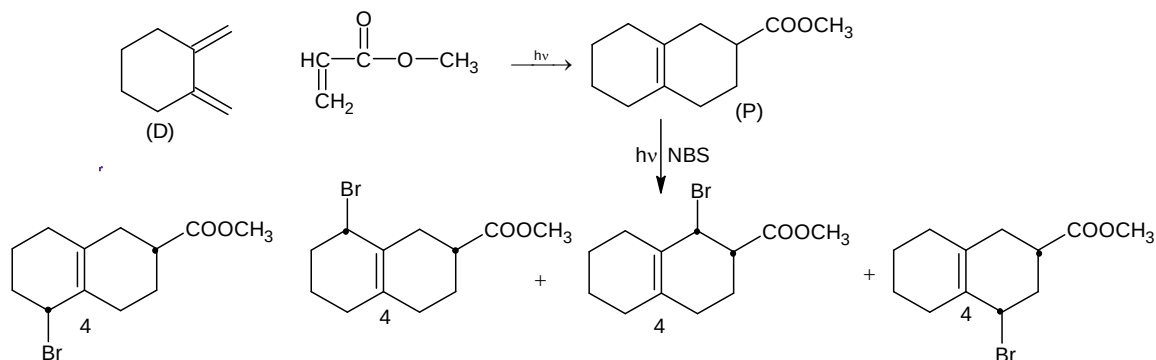
Section – B

30. 2

Sol. Only I and III are optically active.

31. 16

Sol.



Value of x = 16

32. 144

Sol.

$$\frac{P^0 - P_s}{P_s} = \frac{\text{mol of solute}}{\text{mol of solvent}}$$

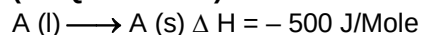
$$\frac{0.13 - 0.12}{0.12} = \frac{\frac{6}{M}}{\frac{46}{92}} = 144$$

Section - C

33. 500.00

34. 400.00

Sol. (for Q.33 and 34)



$$\Delta H = \Delta U + \Delta(PV)$$

$$\Delta H = \Delta U + P(\Delta V)$$

$$= \Delta U + 1 \times 100 \times (-20 \times 50) / 1000$$

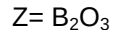
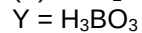
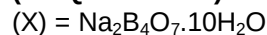
$$\Delta U = -400 \text{ J/Mole}$$

$$|\Delta U| = 400 \text{ Ans.}$$

35. 33.33

36. 68.57

Sol. (for Q.35 and 36)



37. 1.73

38. 3.73

Sol. (for Q. 37 and 38)

y and x are respectively

$$\frac{9 + 3\sqrt{3}}{2Z}, \frac{9 - 3\sqrt{3}}{2Z}$$

Mathematics

PART – III

Section – A

39. D

$$\text{Sol. } \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}^n = \left(3I + 2 \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix} \right)^n = 3^n I + {}^nC_1 3^{n-1} 2A + {}^nC_2 3^{n-2} 2^3 I + {}^nC_3 3^{n-3} (2A)^3 + {}^nC_4 3^{n-4} 2^6 I + \dots$$

$$\text{So, } \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}^n + \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}^n = 2(3^n + {}^nC_2 3^{n-2} 2^3 + {}^nC_4 3^{n-4} 2^6 + {}^nC_6 3^{n-6} 2^9 + \dots) I$$

$$(2 \sum {}^nC_{2k} 3^{n-2k} 2^{3k}) I = ((3+2\sqrt{2})^n + (3-2\sqrt{2})^n) I$$

40. C

$$\text{Sol. If } \lim_{n \rightarrow \infty} \frac{((n+1)!)^{\frac{1}{n+1}} - (n!)^{\frac{1}{n}}}{(n+1) - n} = \ell \Rightarrow \lim_{n \rightarrow \infty} \frac{(n!)^{\frac{1}{n}}}{n} = \ell = \frac{1}{e}$$

$$\lim_{n \rightarrow \infty} ((n+2)!)^{\frac{1}{n+2}} - (n+1)^{\frac{1}{n+1}} + ((n+1)!)^{\frac{1}{n+1}} - (n!)^{\frac{1}{n}} = \frac{2}{e}$$

41. A

$$\text{Sol. } \int f(x)^2 f'(x) dx = \int f(3x)^2 f'(3x) d(3x)$$

$$f(3x)^3 = f(x)^3 + c$$

$$\left\{ \begin{array}{l} f(x)^3 - f\left(\frac{x}{3}\right)^3 = c \\ f\left(\frac{x}{3}\right)^3 - f\left(\frac{x}{3^2}\right)^3 = c \\ \text{or } f\left(\frac{x}{3^2}\right)^3 - f\left(\frac{x}{3^3}\right)^3 = c \\ f\left(\frac{x}{3^{n-1}}\right)^3 - f\left(\frac{x}{3^n}\right)^3 = c \end{array} \right.$$

$$\text{Adding, we get } \lim_{n \rightarrow \infty} \left(f(x)^3 - f\left(\frac{x}{3^{n-1}}\right)^3 \right) = (n+1)c$$

$$f(x)^3 - f(0)^3 = \lim_{n \rightarrow \infty} (n+1)c$$

As L.H.S is bounded $\Rightarrow c = 0 \Rightarrow f(x) = f(0)$; $f(x)$ is constant

$$f(x) = 2; (x-1)^2 + (\cos x - 1/2)^2 + \frac{3}{4} = 0$$

$$\cos^2 x - \cos x + 1 + x^2 - 2x + 1 = 0 \text{ no solution}$$

42. D

$$\text{Sol. As } \omega_1 = \frac{2z_2 z_3}{z_2 + z_3}, \omega_2 = \frac{2z_1 z_3}{z_1 + z_3}, \omega_3 = \frac{2z_1 z_2}{z_1 + z_2}$$

$$\text{The required equation is } (ab-1)z^3 + 2(b^2+a)z^2 + 8bz + 8 = 0$$

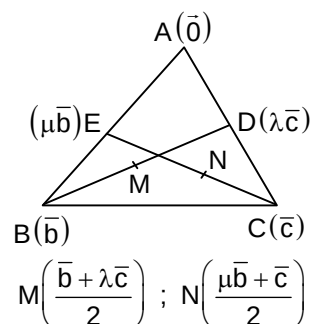
43. A, B, C

Sol. $\Delta = \frac{1}{2} |\vec{b} \times \vec{c}|$; $\Delta_1 = \frac{(1-\mu\lambda) |\vec{b} \times \vec{c}|}{8}$

$$\Delta_2 = \frac{1}{2} \mu\lambda |\vec{b} \times \vec{c}|$$

$$\Delta_3 = \frac{1}{2} (1-\mu\lambda) |\vec{b} \times \vec{c}|$$

$$\Delta_4 = \frac{(1-\mu\lambda) |\vec{b} \times \vec{c}|}{8}$$



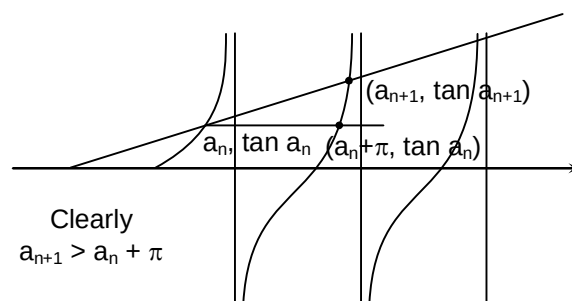
44. A, B, D

Sol. $\frac{dy}{dx} = \cos x \Big|_{x=a_n} = \frac{\sin a_n}{a_n + \frac{\pi}{2}}$

$$a_n + \frac{\pi}{2} = \tan a_n$$

Taking intersection of $y = \tan x$

$$\text{and } y = x + \frac{\pi}{2}$$



45. A, B, C

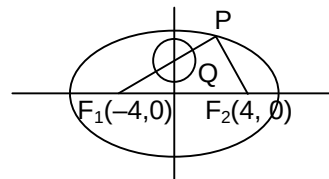
Sol. $PQ + PF_2 = 2a - QF_1$

$$(PQ + PF_2)_{\max} = 12 - (QF_1)_{\min} = 12 - (4) = 8$$

$$(PQ + PF_2)_{\min} = 12 - (QF_1)_{\max} = 12 - 6 = 6$$

For option C, PF_1 is tangent to circle

$$\text{So } QF_1 = \sqrt{24} \text{ so } PQ + PF_2 = 12 - \sqrt{24}$$



46. A, D

Sol. Let p is number of factors of m which are less than equal to 6, out of these k are odd factors $P(A \cap B) = P(A) \cdot P(B)$ (as A, B are independent)

$$\frac{k}{6} = \frac{1}{2} \cdot \frac{p}{6} \Rightarrow p = 2k$$

Case-I : $k = 1$, $p = 2$, possible numbers 2, 2×7 , 2×11 , 2×13 , 2×17 , 2×19 , 2×23 , 2×29 **Case-II** : $k = 2$, $p = 4$, possible numbers 6, 6×7 , 20**Case-III** : $k = 3$, $p = 6$, possible number 60

47. A, B, C

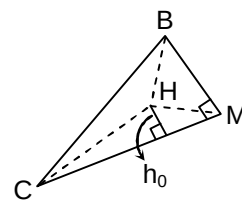
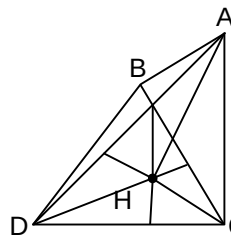
Sol. Let $\perp$ distance from H to BC , BD and CD be p , $2p$, $3p$

$$\frac{1}{2} \times 12(p + 2p + 3p) = \frac{1}{2} 12^2 \sin 60^\circ$$

$$p = \sqrt{3}$$

$$\frac{1}{2} \times 6 \times 2\sqrt{3} + \frac{1}{2} \times 12\sqrt{3} + \frac{1}{2} \times h_0 \times 6\sqrt{3}$$

$$= \frac{1}{2} \left(\frac{1}{2} \times 12^2 \sin 60^\circ \right) \therefore h_0 = 2. \text{ So } \perp \text{ distance of } A \text{ from } CM = \sqrt{2^2 + 3^2} = \sqrt{13}$$



48. B, C, D

$$\text{Sol. Let } f(x) = \begin{cases} 1 & 0 < x < a \\ -1 & a < x < b \\ 1 & b < x < 1 \end{cases}$$

$$\text{As } \int_0^1 f(x) dx = 0 \Rightarrow a - (b - a) + (1 - b) = 0 ; 2a + 1 = 2b$$

$$\int_0^1 x f(x) dx = \int_0^a x f(x) dx + \int_a^b x f(x) dx + \int_b^1 x f(x) dx = \frac{a^2}{2} - \left(\frac{b^2 - a^2}{2} \right) + \frac{1 - b^2}{2} = \frac{2a^2 - 2b^2 + 1}{2}$$

$$I = \frac{2a^2 - 2\left(\frac{2a+1}{2}\right)^2 + 1}{2} = \frac{1-4a}{4} ; I_{\max} = \frac{1}{4} \text{ similarly minimum is } -\frac{1}{4}$$

Section - B

49. 77

$$\text{Sol. } \sqrt{x} = t$$

$$y = t^3 + \frac{1}{8t^3} - \frac{17}{4} \left(t^2 + \frac{1}{4t^2} \right)$$

$$y = \left(t + \frac{1}{2t} \right)^3 - 3 \left(t + \frac{1}{4t} \right) - \frac{17}{4} \left(\left(t + \frac{1}{2t} \right)^2 - 1 \right)$$

$$y = u^3 - \frac{3}{2}(u) - \frac{17}{4}(u^2 - 1) \left(\text{where } u = t + \frac{1}{2t} \geq \sqrt{2} \right) \text{ which has minimum at } u = 3$$

$$y(3) = 27 - \frac{9}{2} - \frac{17}{4}(8) = -\frac{23}{2}$$

50. 1

$$\text{Sol. } q = \sum_{r=1}^{\infty} \frac{1}{2^r} \left(1 - \frac{1}{2^r} \right)^{2021} = \sum_{r=1}^{\infty} \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{2^{(k+1)r}}$$

$$\sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \sum_{r=1}^{\infty} \frac{1}{2^{(k+1)r}} = \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{(2^{k+1} - 1)} = 1 - \sum_{k=1}^{2021} \frac{(-1)^{k-1} {}^{2021}C_k}{2^{k+1} - 1}$$

$$q = 1 - p$$

51. 216

Sol. So the graph of $y = f(x)$ will touch line $y = M$ at $x = \alpha$ and $x = \beta$

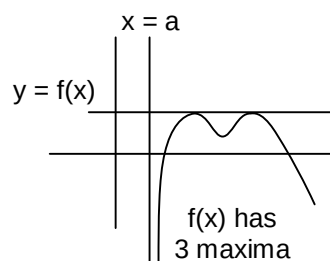
$$\text{so } g(x) = M(x - \alpha) - (x - \alpha)^2(x - \beta)^2$$

$$f(x) = M - \frac{(x - \alpha)^2(x - \beta)^2}{(x - \alpha)}$$

$$g'(x) = M - 2(x - \alpha)(x - \beta)(2x - (\alpha + \beta))$$

$$= M - 4 \left(\left(x - \left(\frac{\alpha + \beta}{2} \right) \right)^2 - \left(\frac{\alpha - \beta}{2} \right)^2 \right) \left(x - \left(\frac{\alpha + \beta}{2} \right) \right)$$

So $g'(x) = 0$ should not have 3 distinct roots



$$g''(x) = -4 \left(3 \left(x - \left(\frac{\alpha + \beta}{2} \right) \right)^2 - \left(\frac{\alpha - \beta}{2} \right)^2 \right) = -12 \left(\left(x - \left(\frac{\alpha + \beta}{2} \right) \right)^2 - 9 \right)$$

$$g' \left(\left(\frac{\alpha + \beta}{2} \right) + 3 \right) g' \left(\left(\frac{\alpha + \beta}{2} \right) - 3 \right) \geq 0$$

$$(M - 216)(M + 216) \geq 0 \Rightarrow M \geq 216$$

Section - C

52. 65.00

53. 25.00

Sol. (for Q. 52.-53)

Roots of the equation $x^2 - (a + 5)x + 2a = 0$ are integers

$$\alpha + \beta = a + 5; \alpha\beta = 2a$$

$$\Rightarrow (\alpha - 2)(\beta - 2) = -6 \text{ all combinations}$$

For (α, β) are $(-4, 3), (-1, 4), (0, 5), (1, 8)$

54. 882.00

Sol. For n black and n white balls number of cases are a_n so $a_n = a_{n-1} + a_{n-2}$ and $a_1 = 1, a_2 = 2, a_7 = 21$

55. 595.00

Sol. In this case if for n black and n white balls total cases are b_n , then $b_n = b_{n-1} + 2b_{n-2}$
 $b_1 = 1; b_2 = 3; b_7 = 85$

56. 4.00

57. 2.66

Sol. Equation of AB $y = \frac{2t}{t^2 - 1}(x - 1)$

$$\text{As } AB \perp CF \Rightarrow \left(\frac{2t_1}{t_1^2 - 1} \right) \left(\frac{2t}{t^2 - 1} \right) = -1$$

$$\text{As } k = 2t_1 \text{ and } k = \frac{2t}{t^2 - 1}(h - 1)$$

$$\text{We get, } 4y^2 + (x - 1)(y^2 - 4) = 0$$

$$\text{So } y = g(x) \text{ is } 4x^2 + (y - 1)(x^2 - 4) = 0$$

$$\Rightarrow g(x) = \frac{3x^2 + 4}{(4 - x^2)}$$

